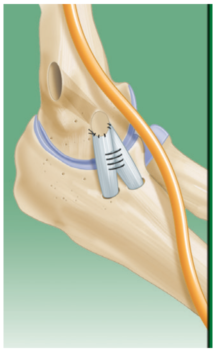
61



Elbow Tendinopathies and Bursitis

Jennifer Moriatis Wolf

Lateral and medial epicondylitis are common elbow tendinopathies involving tendon origins that most commonly present in middle age. These are typically self-limited, with a multitude of conservative and operative options described. In contrast, biceps and triceps tendinitis are insertional tendinopathies. Finally, olecranon bursitis is an inflammation of the dorsal bursa at the elbow. This chapter will evaluate the literature and current evidence about diagnostic maneuvers and treatment options to allow the sports physician to effectively treat patients with these problems.

LATERAL AND MEDIAL EPICONDYLITIS

History

Initially described secondary to lawn tennis,[1] lateral epicondylitis affects approximately 3% of the population during some point in their lives.[2,3] It is far more common than medial epicondylitis, with a 3 to 6 : 1 ratio.[4] Patients with lateral and medial epicondylitis present with complaints of pain on the respective side of the elbow. Also known as tennis elbow (lateral) or golfer's elbow (medial), these entities present most commonly in persons between 30 and 50 years of age.[5] Men and women are equally affected, with a higher occurrence rate in non-athletes.[6]

Symptoms are typically insidious in onset, although some patients develop complaints after a traumatic incident. The pathophysiology of lateral epicondylitis has been shown to involve the origin of the extensor carpi radialis brevis (ECRB) tendon, with infiltration of vascular buds and fibroblast proliferation termed "angiofibroblastic hyperplasia."[7] Another histologic evaluation noted vascular proliferation and hyaline degeneration, without an inflammatory response.[8] Occasionally, the extensor digitorum communis and extensor carpi radialis longus may also be involved. The most common presenting complaint is pain with lifting. As many activities of daily lifting are performed with the forearm in neutral rotation, lifting in this position requires slight wrist extension, which may cause overuse or microtears of the ECRB tendon origin as the source of pain.[9] Nirschl et al. classified lateral epicondylitis by pathologic and pain phases as a guide to the severity of the problem (Table 61.1).[10]

Medial epicondylitis is less common and is not well understood. Patients complain of an insidious onset of pain at the medial aspect of the elbow. Risk factors for medial epicondylitis are similar to those of lateral epicondylitis and include repetitive overuse and obesity. At surgery, Gabel and Morrey observed a nidus of granulation tissue at the flexor-pronator origin overlying the pronator teres and flexor carpi radialis origin.[11] They noted that this entity often overlaps with symptomatic ulnar nerve compression.

Physical Examination

On examination of the patient with tennis elbow, patients are typically tender at or just distal to the lateral epicondyle with palpation. The most characteristic examination maneuver is the provocation of pain at the lateral elbow with resisted wrist extension of the patient's fist, forearm pronated and the elbow fully extended (Fig. 61.1).[12] The Cozen test is a similar maneuver, except that the examiner grasps the elbow to stabilize it while asking the patient to extend the clenched fist against resistance.[12] To rule out posterolateral rotatory instability due to lateral ulnar collateral ligament injury, the posterolateral rotatory drawer test is performed. This is done with the patient supine, with the arm overhead and held externally rotated; the examiner stabilizes the humerus with one hand and applies a posterolateral rotatory stress by grasping the forearm and attempting to push the radial head posteriorly.[13]

At the medial elbow, patients have tenderness at the flexor-pronator origin at the medial epicondyle. Pain with resisted forearm pronation is felt to be the most sensitive test for medial epicondylitis,[11] while pain with resisted wrist flexion may also be indicative. Because of the overlap of medial epicondylitis and ulnar nerve compression, Tinel's testing over the ulnar nerve posterior to the medial epicondyle should be performed. Gabel and Morrey classified medial epicondylitis by the degree of ulnar nerve involvement, as follows: type IA with no symptoms of ulnar nerve compression; type IB with mild ulnar nerve symptoms; and type II with moderate or severe ulnar nerve compression in association with medial epicondylitis.[11] The medial collateral ligament should be examined to rule out instability with valgus load applied to the elbow at 30 degrees of elbow flexion.

Differential diagnosis of these conditions includes radial tunnel syndrome, characterized by aching pain in the proximal forearm distal to the lateral epicondyle with many similar positive provocative tests,[9] medial triceps tendinitis or snapping,[14] and injury or sprain to the lateral and medial collateral ligaments. In addition, neurologic conditions such as pronator syndrome and cubital tunnel syndrome as well as elbow pathologies such as plica, osteochondral defects, and joint degeneration need to be considered.

Abstract

Epicondylitis and tendinitis of the elbow are common entities, both with options for extensive conservative treatment. Both entities represent tendinopathy, or degenerative change in tendon structure, more than true tendon inflammation. Understanding this distinction is important to choose appropriate treatment options, including therapy, injections, and surgery. Olecranon bursitis can be septic or nonseptic and these diseases are treated differently.

Keywords

tennis elbow
lateral epicondylitis
medial epicondylitis
olecranon bursitis
triceps tendinitis
biceps tendinitis

TABLE 61.1 Nirschl Classification of Pathology and Pain in Tennis Elbow

Pathology Staging Score	Pain Phase Score
I. Temporary irritation	1. Pain after exercise only, lasting <24 hours before resolution
	2. Pain after exercise, lasting >48 hours, resolves with warming up
II. Permanent tendinosis <50% cross-sectional area	3. Pain with exercise not affecting the activity
III. Permanent tendinosis >50% cross-sectional area	4. Pain with exercise that alters the sports activity
	5. Pain caused with heavy activities of daily living (ADLs)
	6. Intermittent pain at rest, and pain with light ADLs
IV. Partial or complete ECRB tendon rupture	7. Constant pain at rest, pain that wakes patient up from sleep

Modified from Nirschl RP, Ashman ES. Elbow tendinopathy: tennis elbow. *Clin Sports Med.* 2003;22(4):813–836.

Fig. 61.1 Resisted wrist extension test for lateral epicondylitis.

Imaging

Tennis and golfer's elbow are primarily clinical diagnoses, and the role of imaging is to rule out other conditions about the elbow. Plain radiographs are typically normal, with faint calcifications occasionally seen at the tendon origin (Fig. 61.2). Pomerance evaluated 294 radiographs in patients diagnosed with lateral epicondylitis and noted that 7% had this finding; however, radiographic findings changed management only twice, for patients noted to have osteochondritis dissecans who went on to surgical treatment for this diagnosis.[15]

Ultrasound has been used to evaluate the tendon origin, with findings of decreased echogenicity and peritendinous soft tissue thickening. Abnormalities of the deep fibers of the ECRB may also be visualized.[16] A recent systematic review of 15 diagnostic studies confirmed the utility of gray-scale ultrasound in identification of hypoechogenicity and bone changes at the lateral epicondyle (Fig. 61.3).[17] Heales et al. reported a case-control comparison of ultrasound of patients with lateral epicondylitis, interpreted by blinded radiologists, and noted a sensitivity of 90% and specificity of 47% when greyscale and power Doppler imaging were combined.[18] Magnetic resonance imaging (MRI) may show signal changes at the ECRB origin on T1-weighted images, with fluid or thinning seen at the tendon origin on T2 images, interpreted as enthesopathy or partial tear (Fig. 61.4).[19] van Kollenburg et al. noted that patients with lateral epicondylitis were significantly more likely to have signal changes interpreted as ECRB tendinosis on MRI than controls ($P < .001$), but that both groups had similar proportions of signal changes in the lateral collateral ligament and reading of partial ECRB defect.[19] The same group also evaluated MRI reports of 3374 patients, noting that signal change in the ECRB was common in both symptomatic and asymptomatic elbows, and the incidence of signal change increased over time with increasing age.[20] A recent study suggests that the role of MRI is not for diagnostic or prognostic information but rather to rule out pathology in equivocal or recalcitrant cases of epicondylitis.[21]

Fig. 61.2 Faint calcifications adjacent to the lateral epicondyle seen on plain radiograph.

Fig. 61.3 Ultrasound findings in lateral epicondylitis. Arrowheads demonstrate enlargement and hypoechoic areas (black areas) within the proximal common extensor tendon compatible with tendinosis. (Courtesy Leonardo Oliveira, MD.)

Fig. 61.4 Typical findings on magnetic resonance imaging with signal change in the extensor tendon origin.

Decision-Making Principles

Because the treatment of epicondylitis is primarily nonsurgical, decision-making is based on a discussion with the patient about the evidence regarding various treatment options. It is important for the patient to know that no matter what treatment is chosen, epicondylitis often takes a long time to resolve. Given that no one particular treatment method or algorithm has universal acceptance or been deemed superior to another in these conditions, patients are often initially managed nonoperatively with special considerations given to the individual's vocational and avocational activities. Avoidance of a known aggravating activity is likely where treatment should begin. The remaining methods of treatment are designed to alleviate or reduce symptoms, although their effect may not be immediate or curative. Surgery for epicondylitis is considered as a last resort after other treatments have failed. Although many retrospective studies have reported good outcomes for surgery, more recent meta-analyses have not been able to support a definitive role for surgery for epicondylitis, as the outcomes can be unpredictable.[22]

Treatment Options

There is increasing evidence that lateral and medial epicondylitis are self-limited entities that resolve on their own over time. Smidt et al. performed a randomized controlled trial comparing corticosteroid injection, physiotherapy, or a wait-and-see approach in 185 patients in the Netherlands. At 1 year, success rates were 69% for injections, 91% for therapy, and 83% for wait-and-see, with no significant differences between therapy and observation (see Fig. 61.2).[23] Szabo has recommended reassurance that continued use of the arm will not damage or worsen the tendinosis, and emphasis of the self-limited nature of lateral or medial epicondylitis.[24]

Splinting

The use of a forearm support band or tennis elbow strap has been proposed to reduce stress at the ECRB tendon origin. Meyer et al. noted in a cadaver study that increased band compression caused concomitant force reduction at the ECRB origin.[25] Others noted that wearing a support band increased the rate of fatigue of the wrist extensors, and recommended against their use.[26] A recent randomized trial comparing a forearm band and extensor strengthening exercises showed that both groups improved with time, with no differences between them.[27]

The use of wrist splinting has also been described, with the hypothesis of decreasing the stretch and use of the ECRB. Van de Streek compared a forearm splint to elbow band treatment in 43 patients in a randomized trial and noted no differences in grip or pain after 6 weeks, concluding that both showed some effect.[28] Another randomized trial in 42 patients comparing wrist extension splinting to counterforce strap noted significantly better pain relief with the splint, although there was no difference in measures of function, including the Mayo Elbow Performance score or American Shoulder and Elbow Society (ASES) scores.[29]

Therapy

The use of deep massage, eccentric exercises, strengthening, ultrasound, and iontophoresis have all been proposed as part of therapeutic interventions for epicondylitis; however, there is limited evidence to support one as superior. In Smidt's study comparing therapy, injection, and observation, physiotherapy had the highest success rate at 1 year.[23] The modality of friction massage was described by Cyriax to relieve pain and increase blood flow.[30] A recent randomized trial showed superior results with a supervised exercise program compared to Cyriax physiotherapy for tennis elbow, although both groups improved with respect to pain and function.[31] A meta-analysis evaluating exercise in the treatment of epicondylitis showed that resistance exercises resulted in improvement in tennis elbow symptoms, with eccentric stretching being the most studied.[32] A randomized trial comparing eccentric to concentric exercise showed the largest decrease in pain when eccentric and concentric contractions were combined with isometric exercises.[33]

Ultrasound is thought to provide benefit through deep heating of tissues. Binder et al. noted a significant difference in lateral epicondylitis patients treated with ultrasound compared to placebo treatment;[34] however, later randomized trials have shown no differences between ultrasound and sham-ultrasound treated groups.[35,36] A recent randomized trial comparing exercise and therapeutic ultrasound to corticosteroid injection in 49 subjects demonstrated significant improvement in pain and function in the exercise group at 12 weeks.[37]

Iontophoresis delivers water-soluble medications such as dexamethasone or saline through the skin with the use of electrical current. Stefanou et al. noted short-term benefit of dexamethasone iontophoresis in a randomized study with comparison to injection of dexamethasone.[38] In a double-blind comparison study, however, Runeson and Haker noted no significant differences in 64 patients randomized to iontophoresis or placebo.[39]

Injections

Multiple types of injections have been used for the treatment of epicondylitis. The most common is corticosteroid injection, which has been shown in systematic reviews to have a short-term effect on pain relief for up to 6 weeks (see Fig. 61.3).[40] Given that epicondylitis is a noninflammatory condition, the mechanism of action of steroid injection is uncertain. Randomized trials and a recent meta-analysis of corticosteroid versus placebo injection for tennis elbow showed no differences in outcomes as measured by either Disabilities of the Arm, Shoulder, and Hand (DASH) or pain scores.[41,42]

There has been recent interest in injections of whole blood and platelet-rich plasma (PRP), with the purported mechanism of stimulating reversal of the angiofibroblastic tendinosis by the delivery of humoral mediators and growth factors.[43] Whole or autologous blood contains these factors, and separating blood components by centrifugation to inject plasma concentrates the solution. Edwards and Calandruccio initially described pain relief and functional improvement in two-thirds of a group with chronic lateral epicondylitis treated with one or two autologous blood injections in a retrospective study.[44] Studies of PRP in lateral epicondylitis have shown mixed results, with some randomized trials showing significantly better results compared to corticosteroid,[45,46] while others demonstrated no differences among PRP, corticosteroids, or placebo saline.[47,48] A recent systematic review and network analysis comparing injections of autologous blood to PRP showed comparable outcomes, with both injection types being superior to steroid injections, with a higher risk of complications with autologous blood injection.[49]

Less commonly used injections include botulinum toxin and prolotherapy. Botulinum works by temporary paralysis of the extensor origin, and limited studies have shown effective results in decreasing pain. A randomized multicenter trial showed significantly better results with botulinum injection compared to placebo at 18 weeks, although middle finger extension was weakened by the botulinum temporarily.[50] A different formulation, prolotherapy, consists of either hypertonic glucose or saline injected into the ECRB origin in an effort to sclerose pathologic neovascularization and provide a toxic effect on granulation tissue.[51] While evidence is limited, a recent randomized trial comparing prolotherapy to corticosteroid showed improvement in both groups without significant differences between them.[52]

Surgical Treatment

There are several surgery options for epicondylitis, typically reserved for patients who have failed extensive nonoperative treatment, which include open or arthroscopic débridement, extensor release or repair, extensor repair, and denervation. The classic surgical approach to lateral epicondylitis is termed the "Nirschl procedure" and involves open débridement of all identifiable granulation and/or fibrous tissue at the ECRB origin, (Fig. 61.5) followed by decortication or drilling of the lateral epicondyle prominence to improve blood supply.[7] The overlying fascial edges of the extensor carpi radialis longus and extensor digitorum communis are closed.

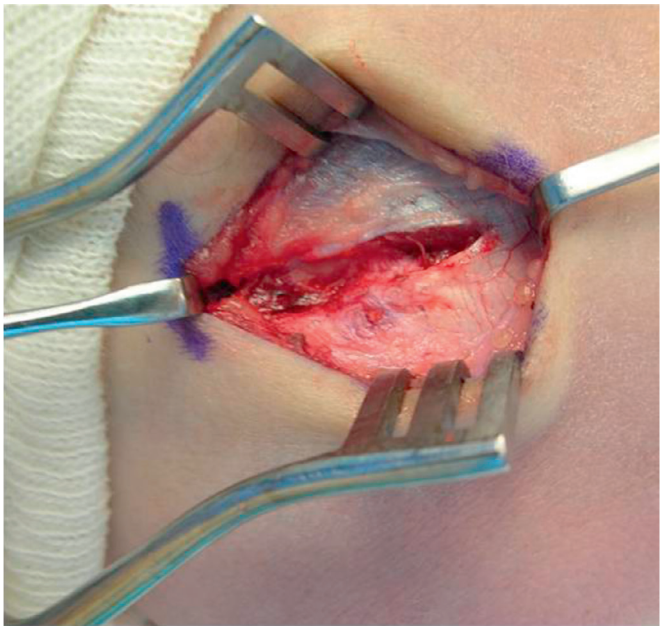


Fig. 61.5 Open surgical débridement for epicondylitis (the Nirschl procedure).

Other authors have described percutaneous[53] or open[54] release of the extensor origin down to the capsule of the elbow, after which the extensors are allowed to retract distally. The extensor muscular origin is attached at multiple points to the surrounding fascial bands and to underlying joint capsule, and thus extensor weakness is not observed after this procedure. A recently described alternative is the use of ultrasound-guided instrumentation to perform microresection and tenotomy at the tendon origin. A retrospectively evaluated cohort of 20 patients showed significant reduction in pain and improved function at 3-year followup.[55]

An alternate procedure involves débridement of devitalized tissue and repair of the ECRB to the lateral epicondyle using suture anchors[56] or bone tunnels.[57] Additionally, denervation of the lateral elbow has been described by Kaplan[58] and others,[59,60] with careful dissection and division of multiple cutaneous radial nerve branches while preserving the posterior interosseous nerve (PIN).

Arthroscopic management of lateral epicondylitis has been demonstrated, with débridement and release of the underside origin of the ECRB. The proposed advantages of arthroscopy are that it is minimally invasive and is able to manage concomitant intra-articular pathology, which has been seen in up to one-third of patients.[61] It is only necessary to access the anterior compartment of the elbow, with use of a modified anterolateral portal established using an inside-out technique, starting 2 to 3 cm proximal and anterior to the lateral epicondyle.[62] Synovitis, identified at the tendon origin and lateral capsule, is débrided using a shaver. The ECRB is then released using monopolar thermal dissection or a shaver, taking care to limit the release to the area anterior to the midline of the radiocapitellar joint in order to preserve the lateral collateral ligament origin. Some surgeons will drill the epicondyle, or débride the exposed underside using an arthroscopic burr.

At the medial epicondyle, it is necessary to first identify and protect the ulnar nerve. The nerve is typically decompressed in situ through a mini-incision, and a limited débridement of the underside of the flexor pronator origin is performed using a shaver. During débridement, the ulnar nerve is retracted medially to protect it.

Author's Preferred Technique

My management of lateral or medial epicondylitis begins with a discussion of the evidence for and against the multiple treatment options. I emphasize the typical self-limited nature of the condition and caution that this tendinosis may remain symptomatic up to 1 to 2 years before resolving, except in a small number of people. I also reassure patients that they are not causing themselves harm or damage by using the arm, and that they are not causing joint instability or tissue damage that will lead to disability. For each patient, I demonstrate eccentric stretching exercises for tendinosis,[32] and discuss that this expectant treatment alone can be sufficient to treat epicondylitis.

I offer referral to therapy and discuss injection options. I always emphasize the limited course of pain relief from steroid injections, approximately 6 weeks in length, but will give one at a patient's request. For patients presenting with greater than 6 months of symptoms and especially with chronic epicondylitis, I will offer an autologous blood injection, as my anecdotal experience has shown that this work bests in that patient group.

For surgical management of lateral epicondylitis, I perform the Nirschl procedure with open débridement of the ECRB origin. This is typically done under Bier block anesthesia, with a longitudinal incision over the lateral epicondyle and ECRB origin. The fascia is incised with retraction of the extensor carpi radialis longus and extensor digitorum communis, exposing the deep ECRB. The origin is débrided of any gray, shiny tissue to leave healthy tendon behind. I typically do not open the elbow joint. I use a rongeur or osteotome to decorticate the prominence of the lateral epicondyle, taking care not to go too inferiorly to protect the origin of the lateral collateral ligament. Sometimes, there may be some degenerative changes on the undersurface of the extensor digitorum communis, which I also débride, as failure to do so may lead to suboptimal results. After irrigation, the fascia of the extensor carpi radialis longus (ECRL) and extensor digitorum communis (EDC) is closed using 2-0 resorbable sutures, and the skin is then closed using a resorbable monocryl subcuticular suture.

For medial epicondylitis, I perform an open débridement of the flexor-pronator origin, focusing on the interval between the pronator teres and the flexor carpi radialis at their attachment to the medial epicondyle. While not typically able to visualize a nidus of abnormal tissue as advocated by Gabel and Morrey, I remove any devitalized tissue at the origin and decorticate the medial epicondyle with a rongeur. Closure and other management are similar to my treatment of lateral epicondylitis.

Postoperative Management

The patient is splinted postoperatively for 2 weeks and then begins therapy for 4 to 6 weeks with a 5-lb lifting limit. Strengthening is begun at 6 weeks, and return to heavy lifting and strenuous sports is not permitted until 3 months.

Return to Play

Resumption of athletic activities is sport-dependent after surgery for lateral or medial epicondylitis. Strenuous or contact sport activity is permitted at 3 months if the patient has tolerated the strengthening phase of rehabilitation. Earlier return, typically at 6 weeks, is allowed for individual sports such as golf, skiing, and swimming, with the elbow taped or supported as necessary.

For nonoperatively treated patients, return to sport is allowed at their tolerance, with the understanding that the elbow may be painful with certain motions or after heavy use.

Results

Open Surgical Management—Lateral Epicondylitis

Dunn et al. reported 10 to 14-year follow-up of the Nirschl open débridement technique in 139 patients in a retrospective review. The authors noted decreased pain scores and improved American Shoulder and Elbow Surgeons' scores, with 84% good to excellent results combining two other functional scales, and 93% of patients who were able to return to sports.[63] A prospective randomized comparative study between percutaneous and open ECRB release showed improvements in both groups, but the percutaneous-release group had significantly better DASH scores, improvement in sports performance, and earlier return to work.[64] Ruch et al. compared open débridement with and without anconeus flap rotation in 57 patients and noted significantly better DASH scores in the group treated with anconeus rotation, but no other differences at 7-month follow-up.[65] A recent Cochrane review of the available evidence for surgical treatment concluded that there was insufficient evidence to support the use of surgical treatment for lateral epicondylitis or the application of one surgical technique as superior.[66]

Arthroscopic Release—Lateral Epicondylitis

Lattermann et al. noted improvement of pain and function scores in a retrospective review of 36 patients treated arthroscopically, although 31% noted mild pain with strenuous activities. Other authors have noted similar positive findings, although heavy laborers and worker's compensation claimants had overall worse outcomes.[67] Szabo et al. compared open, arthroscopic, and percutaneous release of the ECRB in a group of 109 patients with 47.8-month mean follow-up.[68] The authors noted that there was significant improvement in Andrews-Carson scores from pre- to postoperative evaluation, with no differences between groups, and concluded that all three procedures were highly effective in treating lateral epicondylitis. A recent retrospective comparison of arthroscopic to open ECRB débridement showed similar improvement in both groups, but the open procedure cohort had a significantly better pain visual analogue scale (VAS) score during hard or heavy work compared to the arthroscopic group.[69]

Surgical Treatment of Medial Epicondylitis

The majority of reports of treatment for medial epicondylitis involve open débridement of the flexor-pronator origin in patients who have failed lengthy conservative management. Gabel and Morrey noted excellent outcomes in a retrospective review of 26 elbows treated for medial epicondylitis, reporting significantly better results in patients without concomitant ulnar nerve compression neuropathy.[11] A recent retrospective evaluation of 55 patients treated surgically with débridement, medial epicondylectomy, and ulnar nerve release showed significant improvement in pain, grip strength, and DASH and Mayo scores at a minimum

5-year follow-up.[70] Zonno et al. performed a cadaveric study to evaluate the safety of arthroscopic débridement of medial epicondylitis,[71] but this technique has not been reported in clinical studies to date.

Complications

A notable complication of nonoperative treatment is soft tissue atrophy, skin thinning, and hypopigmentation after corticosteroid injections, which is a risk that should be addressed when treating patients with this modality.[72] Subcutaneous injection should be avoided to minimize this complication, and repeated injections increase the risk of atrophy.

Surgical complications include posterolateral rotatory instability if the lateral collateral ligament is damaged in the approach to the common extensors.[73] Kalainov et al. noted this complication after corticosteroid injection in three elbows.[74] The risk of this complication is minimized by taking care not to dissect too far posteriorly on the lateral epicondyle during the surgical approach. Other reported complications are rare, including development of synovial fistulae[75] and transient paresthesias.[76]

Future Considerations

Short-term evidence supports expectant treatment of epicondylitis, with assurance to the patient that this self-limited entity tends to resolve over time. There is a subset of patients, however, who do not resolve their symptoms, eventually leading to surgical treatment. Identification of this group is a focus of continued study. The current evidence for conservative treatment of medial and lateral epicondylitis is strong, with surgery reserved for those who have failed multiple modalities. Potential future considerations include refinement of biologic treatments such as autologous blood and PRP, as well as exploration of the neurologic contributions in patients with chronic epicondylitis.

DISTAL BICEPS AND TRICEPS TENDINITIS

History

Biceps and triceps tendinitis are correctly termed insertional tendinopathies. Similar to epicondylitis and Achilles tendinopathy, they tend to present in middle age with an insidious onset, indicating chronic degenerative changes in aging tendons that do not respond well to overuse.[77] While patients may recall a traumatic event, a careful history will often elicit prodromal aching or pain.

Patients with biceps tendinitis present with anterior elbow pain, which worsens with activities requiring forceful supination (such as application of torque) or elbow flexion.[78] Differential diagnosis includes pronator syndrome, PIN compression, or phlebitis of the brachial vein. Triceps tendinitis is more rarely seen, with patients complaining of posterior elbow soreness, with pain when pushing open doors, or other activities requiring triceps activation without the benefit of gravity.

Physical Examination

In the patient with biceps tendinitis, the most important maneuver is to rule out a complete biceps rupture. This is most effectively done with the biceps hook test, where the examiner has the shoulder abducted and places the patient's forearm in supination with the elbow flexed at 90 degrees, and then hooks his or her finger around the intact biceps tendon from lateral to medial, lifting it slightly anteriorly (Fig. 61.6). O'Driscoll et al. noted that the hook test was more sensitive and specific for biceps ruptures than MRI in a series of surgically treated individuals.[79]

Fig. 61.6 Demonstration of the "hook test" as described by O'Driscoll et al.

Once the biceps tendon is determined to be intact, palpation of the tendon may elicit pain in the patient's tendinopathy, and pulling on the tendon during the hook test should reproduce the symptoms. In addition, pain in the antecubital fossa with resisted forearm supination, more than elbow flexion, is indicative of distal biceps tendinitis.

Patients with triceps tendinitis may have tenderness to palpation over the distal triceps, and often have weakness and pain with resisted extension. The literature is limited on this entity and includes partial tendon tears, which also present with a palpable defect of the broad triceps insertion.[80]

Imaging

Plain radiographs are nondiagnostic and should be ordered only if a bony lesion is being ruled out as a cause of symptoms. MRI is most commonly used to evaluate the biceps and triceps tendon integrity.[81] For the best imaging of the biceps tendon, the FABS position (flexed elbow, abducted shoulder, supinated forearm) has been described to produce a true view of the tendon in the longitudinal plane.[82] Tendinosis is characterized by abnormal signal intensity within the tendon on T2-weighted sequences, as well as tendon thickening, abnormal contour, or swelling on any sequence (Fig. 61.7).[77,82] The triceps tendon is best visualized on sagittal MRI, with fluid-filled defects, signal heterogeneity, and other findings similar to biceps tendinosis at the affected insertion.[81]

Ultrasound has also been used for diagnostic purposes in elbow insertional tendinopathies.[83] Biceps tendinitis may be identified with abnormal fluid signal within the distal tendon. The triceps tendon also shows abnormalities on ultrasound and its integrity and thickness may be evaluated for pathologic differences from normal controls.[84]

Decision-Making Principles

Biceps and triceps tendinopathies are challenging entities to treat, because the patients are often quite symptomatic and it is

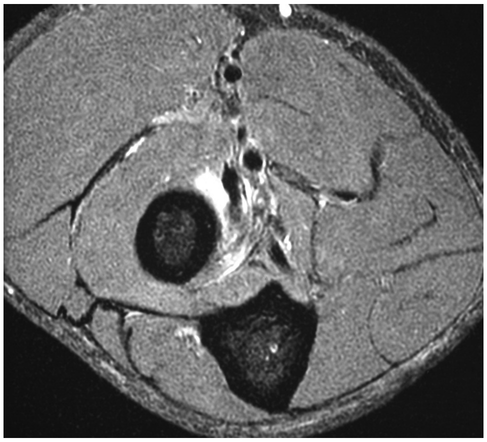


Fig. 61.7 T2-weighted magnetic resonance imaging of biceps tendinosis with fluid in the tendon sheath and high signal surrounding the biceps tendon. (Courtesy George Athwal, MD, FRCSC.)

difficult to modify their elbow use. These insertional tendinopathies are fairly rare, and should be considered typical enthesopathies similar to lateral epicondylitis and Achilles tendinitis when discussing treatment.[77] As such, a trial of nonoperative treatment is always recommended, with assurance of patients that surgery is *not* always necessary. In patients who fail conservative treatment, surgical options have been fairly successful in alleviating symptoms.

Treatment Options and Results

Conservative

Nonoperative treatment of biceps or triceps tendinosis begins with modifying activities that aggravate the patient's symptoms as much as possible. Use of intermittent splinting during heavy loading activities may be considered, with a static elbow splint set between 45 and 90 degrees; however, full-time wear is not recommended due to risks of joint stiffness.[80] Other described conservative measures include nonsteroidal antiinflammatory drugs (NSAIDs) and therapy.[85] More recently, there have been reports of ultrasound-guided PRP for tendinopathy. Sanli reported on 12 patients given a single PRP injection for distal biceps tendinitis using ultrasound, and noted significantly improved pain VAS, biceps strength, and function at 47-month follow-up.[86]

The key in conservative treatment is giving the tendinosis sufficient time, as with epicondylitis, to self-resolve. Mair et al. noted healing of partial triceps ruptures in 6 of 10 professional football players who were treated with time away from competition and protective bracing upon return to play (RTP); one sustained a complete rupture in the splint and the other three had delayed surgery for persistent pain and weakness.[80] Durr et al. reported four patients with partial biceps tendon tears with successful conservative treatment in three-fourths of this group.[87] In general, nonoperative treatment appears to have modest success, possibly associated with the severity of the tendinopathy and demands of the athlete or patient.

Operative

Operative treatment of biceps and triceps tendinosis is indicated when nonoperative treatment has failed. The recommended length of conservative treatment is controversial and varies between 8 weeks[88] and 1 year.[77] The degenerative biceps tendon is débrided through an anterior incision and then repaired using either the same incision with suture anchors[89] or through bone tunnels using a second posterior incision.[90] Kelly et al. described débridement and reattachment all through a single posterior incision.[88] Rokito et al. débrided and reinserted the biceps tendon and reported good subjective outcomes and recovery of strength in a retrospective report.[78] Vardakas et al. evaluated seven patients who failed a mean of 9.5 months of nonoperative treatment and underwent biceps tendon débridement and reattachment through an anterior incision using suture anchors.[85] All were subjectively satisfied at 6 months, had returned to preinjury activity and sport levels, and strength testing showed greater strength than the unaffected side in all cases. Another study evaluated strength at 32 months after surgical débridement and repair of partial biceps tears, showing greater isometric and dynamic elbow flexion strength than the opposite side, although forearm supination strength was 10% weaker.[91]

Triceps ruptures are repaired through bone tunnels in the olecranon, often with graft augmentation from autogenous palmaris longus[92] or hamstring (Fig. 61.8).[93] van Riet et al. reported the results of 23 triceps repairs and reconstructions at 88-month follow-up, noting that triceps strength was on average 82% of the strength in the unaffected arm.[92]

Author's Preferred Technique

Initially, I treat tendinosis or partial tearing of the biceps or triceps insertion conservatively, and discuss with patients that this is similar to other enthesopathies of middle age, which can often resolve without surgical treatment. I use intermittent splinting, therapy, and modification of aggravating activities. I do not recommend corticosteroid injection, as this poses a higher risk of rupture to a degenerative tendon. The implications of biceps or triceps insertional tendon rupture after steroid injection are more concerning than a similar occurrence from injection at the common extensor or flexor origin for epicondylitis, where functional deficits would be less disabling after iatrogenic rupture. I do not have current experience with PRP injection for these tendinopathies.

When patients have failed 6 to 12 months of conservative therapy, I recommend surgical treatment. For biceps tendinitis, I prefer a two-incision technique, with débridement of the tendon and detachment from the anatomic footprint, after which nonresorbable sutures are placed using a locking Krackow configuration in the tendon. The posterior counter-incision is made after passing a curved clamp ulnarly around the bicipital (radial) tuberosity to the dorsal forearm, using the palpable clamp tip for guidance. Using a burr, a trough for the tendon is created at the tuberosity, with drill holes then placed from dorsal to volar to allow passage of the tendon sutures. The elbow is placed in 90 degrees of flexion and fully pronated, and the tendon sutures are tensioned and tied.

For triceps tendinosis, I have performed arthroscopic débridement in one patient, with partial pain relief reported. This was done in the posterior compartment only, with débridement of an acute area of thickened degenerative-appearing tissue using a 3.5 mm shaver. While I have never had to débride and reattach the triceps for tendinosis, this would be done through a straight posterior incision, with similar nonresorbable sutures placed in a locking Krackow technique in the tendon. Obliquely oriented drill holes in the olecranon and proximal ulna are used to pass the sutures, which are then tied over the dorsal surface of the ulna with the elbow flexed at 45 degrees.

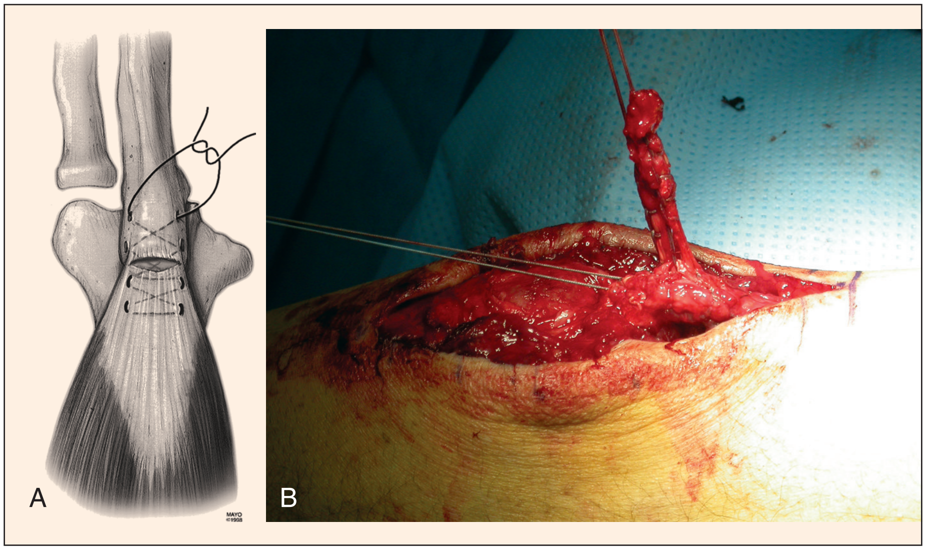


Fig. 61.8 Triceps tendon repair with locking Krackow stitch.

Postoperative Management

After biceps tendon reattachment, the patient is placed in a splint at 90 degrees of elbow flexion with the forearm supinated to decrease tensile forces on the repair. Graded active extension and passive flexion are allowed at 2 to 3 days, and a removable splint in the same position as the immediate postoperative splint is employed and worn between exercises for 4 to 6 weeks. Active elbow flexion and forearm supination are begun at 4 weeks. Strengthening is not begun until 3 months postoperatively.

After triceps tendon surgery, the patient is placed in a splint at 45 degrees of elbow flexion, with graded active flexion and passive extension started within the week. Flexion is slowly increased by 10 to 20 degrees per week. Active extension is begun at 4 to 6 weeks, and similarly to the biceps, strengthening is deferred until 3 months.

In both sets of patients, oral indomethacin is used for one month to prevent heterotopic ossification, unless contraindicated by other co-morbidities. I prefer three times daily dosing, at 25 mg, as opposed to the extended release, as patients seem to tolerate this dosage better.

Return to Play

RTP is allowed as strengthening begins, with protection in a splint initially. Particularly in contact sports, it is critical to defer full participation until the tendon repair is fully healed enough to tolerate heavy use.

Complications

Surgical complications of biceps repair include heterotopic ossification, nerve injuries (median, lateral antebrachial, and PIN), and re-rupture. Heterotopic ossification occurs more frequently with two-incision repairs,[94] but can be seen with both techniques. The worst-case scenario is the development of heterotopic ossification limiting elbow motion or radioulnar synostosis,[95] both of which require secondary surgery. While heterotopic ossification is typically non-bridging, oral indomethacin has been suggested as a prophylactic measure. Anakwenze et al. reported on the outcomes of 34 patients with complete biceps ruptures treated with a two-incision repair and chemoprophylaxis using indomethacin.[96] At a mean of 22 months, the authors noted no heterotopic ossification, reruptures, or synostoses, although they acknowledged the limitation of no comparative control group not treated with indomethacin.

Nerve injuries during biceps repair involve either the lateral antebrachial cutaneous (LABC) nerve or the PIN, both related to excessive retraction in the anterior or single-incision approach. Injury to the LABC is typically neuropraxic, with the patient noting transient numbness on the anterolateral aspect of the forearm. Grewal et al. noted LABC neuropraxia in 19 of 47 patients after a single-incision biceps repair compared to 3 of 43 in the double-incision group, thought due to more extensive retraction time necessary with the single-incision technique.[90] Injury to the PIN, while typically a neuropraxic stretch mechanism as well, is a dreaded complication, causing wrist drop that may take 4 to 5 months to resolve. To decrease the risk of this complication, anterior elbow dissection should be performed with the forearm maximally supinated to move the nerve as far laterally as possible.[97]

Finally, re-rupture may occur, most commonly associated with patient noncompliance, but also potentially due to inadequate fixation methods.[98] This is best prevented by securing the tendon with adequate grasping sutures, using superior tendon-to-bone fixation techniques, and preparing the patient preoperatively for the extent and particular restrictions of postoperative rehabilitation.

Complications of triceps repair include ulnar neuropraxia and rerupture. van Riet noted one case of ulnar neuropraxia, thought likely due to repair of an associated olecranon avulsion with tension-band wiring, and three reruptures after primary repair.[92]

Future Considerations

The natural history of biceps and triceps tendinosis has not been well addressed in the literature, and further prospective studies of these rare entities are needed. Prospective studies could elucidate evidence that supports earlier and more aggressive intervention in these conditions, rather than prolonged and only modestly successful nonoperative management as the standard of care. This is already the prevailing theory for many surgeons, citing that lack of elbow extension and/or forearm supination power may be more debilitating for athletes and manual laborers than the functional deficits engendered by the epicondylitides. Furthermore, protracted conservative care of some individuals may be counterproductive, leading to unnecessary time off work, extended avoidance of competitive sporting activities, and overall frustration on the part of patient and caregiver, when there is little to suggest that these conditions are reliably self-limiting. Detachment and reattachment of the degenerative or partially torn distal biceps tendon, in particular, may reveal more predictable subjective and objective clinical outcomes, especially in high demand patients.

OLECRANON BURSITIS

Olecranon bursitis differs from the other entities described in this chapter in that it is primarily inflammatory in nature. The olecranon bursa is the anatomic "cushion" for the dorsal elbow, which pads the prominent tip of the olecranon (Fig. 61.9). When the subcutaneous bursa becomes inflamed from pressure or trauma, the lining synovial cells produce increased fluid that then distends the bursa. As the bursa is anatomically not connected to the elbow joint, the inflammation is typically limited to the posterior elbow. Olecranon bursitis can be acutely painful, and if contaminated by an open wound due to trauma, may become infected. This section will focus on the evaluation and treatment of nonseptic olecranon bursitis.

History

Olecranon bursitis often presents after an inciting incident, such as a fall directly onto the posterior elbow, other traumatic or overuse mechanism, or repetitive pressure over the dorsal elbow. Patients may complain of pain due to the trauma, but often are minimally symptomatic after a short time passes. They note the presence of a large, sometimes boggy sac at the posterior elbow (see Fig. 61.9). These patients may present with either acute or chronic complaints and to variable types of medical settings apart from the orthopaedic or sports medicine specialist (emergency department, urgent care clinic, or primary care physician office).

Fig. 61.9 Olecranon bursitis with a large mass over the dorsal elbow.

Other contributing diagnoses include gout, calcium pyrophosphate disease (CPPD), and rheumatoid arthritis. All of these comorbid diseases may present with swelling and inflammation of the olecranon bursa, although they are more likely to have separate and often multiple musculoskeletal complaints.

Physical Examination

On inspection, most patients have fullness over the posterior elbow. Acutely, the examiner may sometimes appreciate a fluid wave. In the chronic presentation, the bursal fullness often feels somewhat solid or thickened, reflecting fibrous organization of inflammatory tissue and synovitis. When a small bursa is present, the examiner should palpate the olecranon tip to evaluate for possible spur formation.

Infectious processes more often demonstrate patchy erythema and exquisite pain over the bursa. Loss of elbow motion is a less reliable predictor of infection, since many patients will have concomitant pathology, such as osteoarthritis, or have variable tolerance to stretching of inflamed or swollen soft tissue. The risk of infection is highest in acute presentations, with associated open wounds over the bursa, and in diabetics. A similar presentation, however, may accompany a crystalline deposition disease, which is often deemed a mimicker of infection regardless of the site of involvement. Aspiration, in these instances, may differentiate the two diagnoses and have obvious implications for the natural course and effective treatment of the offending condition.

Imaging

Plain radiographs of the elbow should be obtained to evaluate for olecranon spurs (enthesophytes), as well as calcifications within the bursa (Fig. 61.10). Saini and Canoso compared the radiographs of the affected and unaffected elbow of 28 patients with traumatic olecranon bursitis, and noted that the presence of olecranon spurs and amorphous calcium deposits were higher in bursitis patients (16 of 28) compared to controls (4 of 28).[99] The presence of an olecranon spur is worth noting, because it should be concurrently addressed if surgical treatment is necessary (see Fig. 61.10).

Decision-Making Principles

In nonseptic olecranon bursitis, both conservative and operative options are available. This author's practice does not include aspiration of the bursa, however, because of the risk of converting a nonseptic situation to a septic olecranon bursitis, mandating antibiotics and often inevitable surgical treatment. This view is in contradistinction to that held by many emergency department and primary care physicians.[100] Other accompanying signs and symptoms may lead to a higher clinical suspicion for infection or crystalline deposition disease, where aspiration may be accordingly

Fig. 61.10 Large olecranon spur and associated soft tissue swelling in the bursa.

more justified to differentiate between the two diagnoses. In this regard, the results of aspiration have implications for deciding between management with either anti-inflammatory medication and time or aggressive surgical débridement to prevent progression of an infection to involve the elbow joint. In the rare cases where aspiration is indicated, a direct posterior needle track should be avoided. Rather, the needle should be placed from an anterolateral starting point, through the anconeus or extensors, to prevent the development of a sinus tract.

Treatment Options

Initial treatment of olecranon bursitis is supportive, with compressive wrapping and occasionally elbow splinting in 90 degrees of flexion in the acute setting. Even chronic olecranon bursitis can respond to compressive bandaging.[101] The patient should be informed that compressive wrapping often requires 1 to 2 months of consistent compliance before they see evidence of resolution. In addition, judicious use of anti-inflammatory medication may be considered, particularly in patients with a history of gout.[101]

As noted above, the use of aspiration and corticosteroid injection into the bursa has been advocated. Weinstein et al. reported on aspiration alone versus aspiration and corticosteroid injection for nonseptic bursitis in 47 patients, and noted faster resolution of effusion with the addition of corticosteroid.[102] Complications including infection, skin atrophy, and chronic pain, however, developed in 15 of 25 patients treated with corticosteroid. The risks of corticosteroid injection affecting the underlying triceps should also be considered. van Riet et al. noted that one of the triceps rupture patients in their series had been treated with a steroid injection for olecranon bursitis.[92]

Surgical treatment with excision of the bursa is considered after failure of compressive treatment, or if there is an infectious etiology. Open excision involves complete removal of the bursa along with any olecranon spurring, to prevent recurrence, and

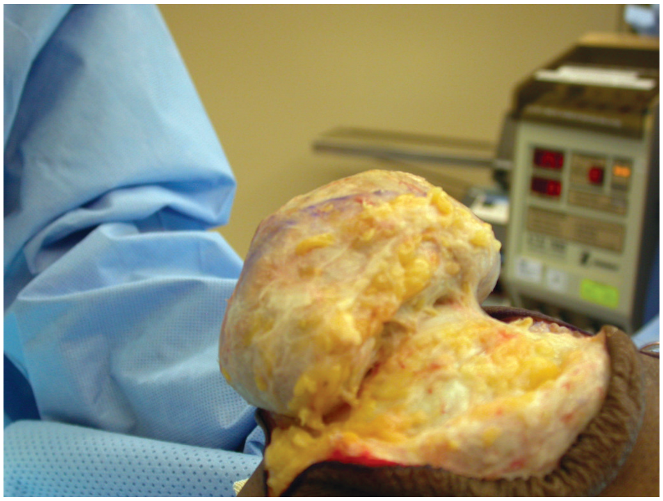


Fig. 61.11 Bursal sac dissected free of skin.

the patient is warned about the minimal soft tissue envelope remaining over the olecranon. This surgery is performed under general or regional block, with the patient positioned supine with a bump under the ipsilateral shoulder and the elbow placed across the chest on a pillow. Use of a sterile blanket in addition is helpful.

A posterior incision with a slight curve to avoid the tip of the olecranon is performed and the bursal sac is carefully dissected free of the skin and subcutaneous fat, which is carefully preserved (Fig. 61.11). The bursa is often firmly adherent to the underlying triceps and needs to be peeled up with sharp dissection. When the bursal sac is removed, the skin edges may need to be trimmed to reduce loose skin at closure, depending on the size of the pathologic bursa.

Arthroscopic excision has also been described, with a posterior elbow arthroscopic setup and use of shavers to débride and remove the bursal sac.[103,104] Rhyou et al. recently reported on the use of arthroscopic bursal resection in 30 patients treated for aseptic and septic olecranon bursitis, noting no recurrences and improved pain and quick DASH scores.[105]

Author's Preferred Technique

When a patient presents with nonseptic olecranon bursitis, I discuss nonoperative treatment first, which is principally full-time compressive bandaging, using either an Ace wrap or neoprene sleeve. The patient may add NSAIDs if they wish, but I believe the primary treatment goal is compression to allow the body to naturally resorb the fluid within the bursa. I emphasize that the compressive bandage should be used as much as possible but may take 1 to 2 months to work. As stated above, I do not recommend aspiration of either septic or nonseptic bursitis. If the bursitis is persistent, this is an indication for surgical excision of the bursa, not serial aspirations.

I perform open surgical excision with the positioning as noted in the treatment section, utilizing a straight posterior incision. After bursal excision, the tourniquet is released and careful hemostasis is achieved in order to leave a dry bed prior to closure. I warn patients about the common reaccumulation of fluid. I also trim the skin to allow a smooth closure without skin redundancy. I then place the patient in a compressive dressing and a splint at 90 degrees. Frequently, compressive bandaging is needed for some time postoperatively.

Postoperative Care

Typically, the patient is splinted for 1 to 2 weeks with the forearm neutral at 90 degrees of elbow flexion and the wrist free. Subsequently, compressive wrapping is recommended while the wound heals further, for the next 2 to 4 weeks. Contact sports and heavy lifting may be resumed at 6 to 8 weeks. Therapy is not usually necessary.

Results

There are limited data regarding surgical bursal excision. Stewart et al. reported retrospectively on 16 patients at the Mayo Clinic treated surgically, and noted 15 of 16 with good results and no recurrence.[106] Other authors noted less positive results in a series of 37 patients treated with bursal excision, with recurrence in eight (22%) and wound healing issues in ten (27%).[107] Arthroscopic resection has shown promising results, with reports of full elbow motion and no recurrence in a group of nine patients.[104]

Complications

Noted surgical complications include recurrence, hematoma, and wound healing problems. Recurrence is poorly defined in the few studies existing, and it is possible this refers to re-accumulation of fluid in the soft tissues as well, since excision of the bursal sac should not allow true recurrence. Degreef and De Smet noted a recurrence in 8 of 37 patients (22%) treated with open bursectomy.[107] Stewart noted postoperative hematoma in one patient, which required surgical drainage.[106]

Wound healing problems have been reported. Degreef and De Smet noted fistula formation, persistent draining wounds, and skin loss requiring flap coverage in 10 of 37 patients reviewed retrospectively.[107] Some of these patients were on chronic anti-coagulation and the cohort included resections of infected bursae, some of whom had open draining wounds preoperatively. Careful management of the soft tissues is required to decrease the risk of these complications.

Future Considerations

Despite the common presentation of olecranon bursitis, there is minimal literature documenting its treatment and clinical outcomes. Further prospective evaluation would be an addition to our current limited knowledge of this entity.

For a complete list of references, go to ExpertConsult.com.

SELECTED READINGS

Citation:
Nirschl RP, Ashman ES. Elbow tendinopathy: tennis elbow. *Clin Sports Med*. 2003;22:813–836.

Level of Evidence:
V, expert opinion

Summary:
This article provides both historical context and the basis of much of the etiology, as well as conservative and operative treatment options for lateral epicondylitis.

Citation:
Smidt N, van der Windt DA, Assendelft WJ, et al. Corticosteroid injections, physiotherapy, or a wait-and-see policy for lateral epicondylitis: a randomised controlled trial. *Lancet*. 2002; 359:657–662.

Level of Evidence:
II

Summary:
This is the definitive support for expectant treatment of epicondylitis as a self-limited disease, and also shows the short-lived effect of steroids as well as the possible impact of therapy.

Citation:
Pierce TP, Issa K, Gilbert BT, et al. A systematic review of tennis elbow surgery: open versus arthroscopic versus percutaneous release of the common extensor origin. *Arthroscopy*. 2017;S0749-8063 (epub ahead of print).

Level of Evidence:
Systematic review

Summary:
This summarized data from 30 articles, and showed that open and arthroscopic techniques were associated with better Disabilities of the Arm, Shoulder, and Hand scores than percutaneous release, with lower pain scores in arthroscopic and percutaneous techniques. There were no overall differences among the three techniques in satisfaction.

Citation:
Hobbs MC, Koch J, Bamberger HB. Distal biceps tendinosis: evidence-based review. *J Hand Surg Am*. 2009;34:1124–1126.

Level of Evidence:
Review, with some Level V expert opinion

Summary:
This article provides a current review of the limited literature on biceps tendinosis and an objective evaluation of treatment options including nonoperative treatment.

Citation:
Sayegh ET, Strauch RJ. Treatment of olecranon bursitis: a systematic review. *Atch Orthop Trauma Surg*. 2014;124(111):1517–1536.

Level of Evidence:
Systematic review

Summary:
This review included 29 studies using primarily Level IV evidence. This showed that nonsurgical treatment is significantly more effective in resolving olecranon bursitis than surgery, and that aseptic bursitis is more difficult to treat than septic bursitis.

REFERENCES

1. Thurston AJ. Conservative and surgical treatment of tennis elbow: a study of outcome. *Aust N Z J Surg*. 1998;68(8): 568–572.
2. Allander E. Prevalence, incidence, and remission rates of some common rheumatic diseases or syndromes. *Scand J Rheumatol*. 1974;3(3):145–153.
3. Sanders TL Jr, Maradit Kremers H, Bryan AJ, et al. The epidemiology and health care burden of tennis elbow: a population-based study. *Am J Sports Med*. 2015;43(5): 1066–1071.
4. Shiri R, Viikari-Juntura E, Varonen H, et al. Prevalence and determinants of lateral and medial epicondylitis: a population study. *Am J Epidemiol*. 2006;164(11):1065–1074.
5. Faro F, Wolf JM. Lateral epicondylitis: review and current concepts. *J Hand Surg Am*. 2007;32(8):1271–1279.
6. Coonrad RW, Hooper WR. Tennis elbow: its course, natural history, conservative and surgical management. *J Bone Joint Surg Am*. 1973;55(6):1177–1182.
7. Nirschl RP, Pettrone FA. Tennis elbow. The surgical treatment of lateral epicondylitis. *J Bone Joint Surg Am*. 1979;61(6A): 832–839.
8. Regan W, Wold LE, Coonrad R, et al. Microscopic histopathology of chronic refractory lateral epicondylitis. *Am J Sports Med*. 1992;20(6):746–749.
9. Scher DL, Wolf JM, Owens BD. Lateral epicondylitis. *Orthopedics*. 2009;32(4).
10. Nirschl RP, Ashman ES. Elbow tendinopathy: tennis elbow. *Clin Sports Med*. 2003;22(4):813–836.
11. Gabel GT, Morrey BF. Operative treatment of medical epicondylitis. Influence of concomitant ulnar neuropathy at the elbow. *J Bone Joint Surg Am*. 1995;77(7):1065–1069.
12. Buckup K. *Clinical Tests for the Musculoskeletal System*. Berlin: Thieme; 2004.
13. Camp CL, Smith J, O'Driscoll SW. Posterolateral Rotatory Instability of the Elbow: Part I. Mechanism of Injury and the Posterolateral Rotatory Drawer Test. *Arthrosc Tech*. 2017;6(2):e401–e405.
14. Rineer CA, Ruch DS. Elbow tendinopathy and tendon ruptures: epicondylitis, biceps and triceps ruptures. *J Hand Surg Am*. 2009;34(3):566–576.
15. Pomerance J. Radiographic analysis of lateral epicondylitis. *J Shoulder Elbow Surg*. 2002;11(2):156–157.
16. Radunovic G, Vlad V, Micu MC, et al. Ultrasound assessment of the elbow. *Med Ultrason*. 2012;14(2):141–146.
17. Dones VC 3rd, Grimmer K, Thoirs K, et al. The diagnostic validity of musculoskeletal ultrasound in lateral epicondylalgia: a systematic review. *BMC Med Imaging*. 2014;14:10.
18. Heales LJ, Broadhurst N, Mellor R, et al. Diagnostic ultrasound imaging for lateral epicondylalgia: a case-control study. *Med Sci Sports Exerc*. 2014;46(11):2070–2076.
19. van Kollenburg JA, Brouwer KM, Jupiter JB, et al. Magnetic resonance imaging signal abnormalities in enthesopathy of the extensor carpi radialis longus origin. *J Hand Surg Am*. 2009;34(6):1094–1098.
20. van Leeuwen WF, Janssen SJ, Ring D, et al. Incidental magnetic resonance imaging signal changes in the extensor carpi radialis brevis origin are more common with age. *J Shoulder Elbow Surg*. 2016;25(7):1175–1181.
21. Walton MJ, Mackie K, Fallon M, et al. The reliability and validity of magnetic resonance imaging in the assessment of chronic lateral epicondylitis. *J Hand Surg Am*. 2011;36(3):475–479.
22. Buchbinder R, Johnston RV, Barnsley L, et al. Surgery for lateral elbow pain. *Cochrane Database Syst Rev*. 2011;(3): CD003525.
23. Smidt N, van der Windt DA, Assendelft WJ, et al. Corticosteroid injections, physiotherapy, or a wait-and-see policy for lateral epicondylitis: a randomised controlled trial. *Lancet*. 2002;359(9307):657–662.
24. Szabo RM. Steroid injection for lateral epicondylitis. *J Hand Surg Am*. 2009;34(2):326–330.
25. Meyer NJ, Pennington W, Haines B, et al. The effect of the forearm support band on forces at the origin of the extensor carpi radialis brevis: a cadaveric study and review of literature. *J Hand Ther*. 2002;15(2):179–184.
26. Knebel PT, Avery DW, Gebhardt TL, et al. Effects of the forearm support band on wrist extensor muscle fatigue. *J Orthop Sports Phys Ther*. 1999;29(11):677–685.
27. Luginbuhl R, Brunner F, Schneeberger AG. No effect of forearm band and extensor strengthening exercises for the treatment of tennis elbow: a prospective randomised study. *Chir Organi Mov*. 2008;91(1):35–40.
28. Van De Streek MD, Van Der Schans CP, De Greef MH, et al. The effect of a forearm/hand splint compared with an elbow band as a treatment for lateral epicondylitis. *Prosthet Orthot Int*. 2004;28(2):183–189.
29. Garg R, Adamson GJ, Dawson PA, et al. A prospective randomized study comparing a forearm strap brace versus a wrist splint for the treatment of lateral epicondylitis. *J Shoulder Elbow Surg*. 2010;19(4):508–512.
30. Stasinopoulos D, Johnson MI. Cyriax physiotherapy for tennis elbow/lateral epicondylitis. *Br J Sports Med*. 2004;38(6): 675–677.
31. Viswas R, Ramachandran R, Korde Anantkumar P. Comparison of effectiveness of supervised exercise program and Cyriax physiotherapy in patients with tennis elbow (lateral epicondylitis): a randomized clinical trial. *Scientific World Journal*. 2012;939645.
32. Raman J, MacDermid JC, Grewal R. Effectiveness of different methods of resistance exercises in lateral epicondylosis–a systematic review. *J Hand Ther*. 2012;25(1):5–25, quiz 26.
33. Stasinopoulos D, Stasinopoulos I. Comparison of effects of eccentric training, eccentric-concentric training, and eccentric-concentric training combined with isometric contraction in the treatment of lateral elbow tendinopathy. *J Hand Ther*. 2017;30(1):13–19.
34. Binder A, Hodge G, Greenwood AM, et al. Is therapeutic ultrasound effective in treating soft tissue lesions? *Br Med J (Clin Res Ed)*. 1985;290(6467):512–514.
35. D'Vaz AP, Ostor AJ, Speed CA, et al. Pulsed low-intensity ultrasound therapy for chronic lateral epicondylitis: a randomized controlled trial. *Rheumatology (Oxford)*. 2006;45(5):566–570.
36. Haker E, Lundeberg T. Pulsed ultrasound treatment in lateral epicondylalgia. *Scand J Rehabil Med*. 1991;23(3):115–118.
37. Murtezani A, Ibraimi Z, Vllasolli TO, et al. Exercise and therapeutic ultrasound compared with corticosteroid injection for chronic lateral epicondylitis: a randomized controlled trial. *Orthop Traumatol Rehabil*. 2015;17(4): 351–357.
38. Stefanou A, Marshall N, Holdan W, et al. A randomized study comparing corticosteroid injection to corticosteroid

iontophoresis for lateral epicondylitis. *J Hand Surg Am*. 2012;37(1):104–109.
39. Runeson L, Haker E. Iontophoresis with cortisone in the treatment of lateral epicondylalgia (tennis elbow)–a double-blind study. *Scand J Med Sci Sports*. 2002;12(3):136–142.
40. Smidt N, Assendelft WJ, van der Windt DA, et al. Corticosteroid injections for lateral epicondylitis: a systematic review. *Pain*. 2002;96(1–2):23–40.
41. Lindenhovius A, Henket M, Gilligan BP, et al. Injection of dexamethasone versus placebo for lateral elbow pain: a prospective, double-blind, randomized clinical trial. *J Hand Surg Am*. 2008;33(6):909–919.
42. Claessen FM, Heesters BA, Chan JJ, et al. A meta-analysis of the effect of corticosteroid injection for enthesopathy of the extensor carpi radialis brevis origin. *J Hand Surg Am*. 2016;41(10):988–998, e982.
43. Kajikawa Y, Morihara T, Sakamoto H, et al. Platelet-rich plasma enhances the initial mobilization of circulation-derived cells for tendon healing. *J Cell Physiol*. 2008;215(3):837–845.
44. Edwards SG, Calandruccio JH. Autologous blood injections for refractory lateral epicondylitis. *J Hand Surg Am*. 2003;28(2): 272–278.
45. Yadav R, Kothari SY, Borah D. Comparison of local injection of platelet rich plasma and corticosteroids in the treatment of lateral epicondylitis of humerus. *J Clin Diagn Res*. 2015;9(7): RC5–RC7.
46. Gosens T, Peerbooms JC, van Laar W, et al. Ongoing positive effect of platelet-rich plasma versus corticosteroid injection in lateral epicondylitis: a double-blind randomized controlled trial with 2-year follow-up. *Am J Sports Med*. 2011;39(6): 1200–1208.
47. Montalvan B, Le Goux P, Klouche S, et al. Inefficacy of ultrasound-guided local injections of autologous conditioned plasma for recent epicondylitis: results of a double-blind placebo-controlled randomized clinical trial with one-year follow-up. *Rheumatology (Oxford)*. 2016;55(2):279–285.
48. Krogh TP, Fredberg U, Stengaard-Pedersen K, et al. Treatment of lateral epicondylitis with platelet-rich plasma, glucocorticoid, or saline: a randomized, double-blind, placebo-controlled trial. *Am J Sports Med*. 2013;41(3): 625–635.
49. Arirachakaran A, Sukthuayat A, Sisayanarane T, et al. Platelet-rich plasma versus autologous blood versus steroid injection in lateral epicondylitis: systematic review and network meta-analysis. *J Orthop Traumatol*. 2016;17(2):101–112.
50. Placzek R, Drescher W, Deuretzbacher G, et al. Treatment of chronic radial epicondylitis with botulinum toxin A. A double-blind, placebo-controlled, randomized multicenter study. *J Bone Joint Surg Am*. 2007;89(2):255–260.
51. Scarpone M, Rabago DP, Zgierska A, et al. The efficacy of prolotherapy for lateral epicondylosis: a pilot study. *Clin J Sport Med*. 2008;18(3):248–254.
52. Carayannopoulos A, Borg-Stein J, Sokolof J, et al. Prolotherapy versus corticosteroid injections for the treatment of lateral epicondylosis: a randomized controlled trial. *PM R*. 2011;3(8): 706–715.
53. Grundberg AB, Dobson JF. Percutaneous release of the common extensor origin for tennis elbow. *Clin Orthop Relat Res*. 2000;376:137–140.
54. Verhaar J, Walenkamp G, Kester A, et al. Lateral extensor release for tennis elbow. A prospective long-term follow-up study. *J Bone Joint Surg Am*. 1993;75(7):1034–1043.
55. Seng C, Mohan PC, Koh SB, et al. Ultrasonic percutaneous tenotomy for recalcitrant lateral elbow tendinopathy: sustainability and sonographic progression at 3 years. *Am J Sports Med*. 2016;44(2):504–510.
56. Thornton SJ, Rogers JR, Prickett WD, et al. Treatment of recalcitrant lateral epicondylitis with suture anchor repair. *Am J Sports Med*. 2005;33(10):1558–1564.
57. Pruzansky ME, Gantsoudes GD, Watters N. Late surgical results of reattachment to bone in repair of chronic lateral epicondylitis. *Am J Orthop (Belle Mead NJ)*. 2009;38(6): 295–299.
58. Kaplan EB. Treatment of tennis elbow (epicondylitis) by denervation. *J Bone Joint Surg Am*. 1959;41-A(1):147–151.
59. Wilhelm A. Tennis elbow: treatment of resistant cases by denervation. *J Hand Surg [Br]*. 1996;21(4):523–533.
60. Rose NE, Forman SK, Dellon AL. Denervation of the lateral humeral epicondyle for treatment of chronic lateral epicondylitis. *J Hand Surg Am*. 2013;38(2):344–349.
61. Lattermann C, Romeo AA, Anbari A, et al. Arthroscopic debridement of the extensor carpi radialis brevis for recalcitrant lateral epicondylitis. *J Shoulder Elbow Surg*. 2010;19(5):651–656.
62. Cohen MS, Romeo AA. Open and arthroscopic management of lateral epicondylitis in the athlete. *Hand Clin*. 2009;25(3):331–338.
63. Dunn JH, Kim JJ, Davis L, et al. Ten- to 14-year follow-up of the Nirschl surgical technique for lateral epicondylitis. *Am J Sports Med*. 2008;36(2):261–266.
64. Dunkow PD, Jatti M, Muddu BN. A comparison of open and percutaneous techniques in the surgical treatment of tennis elbow. *J Bone Joint Surg Br*. 2004;86(5):701–704.
65. Ruch DS, Orr SB, Richard MJ, et al. A comparison of debridement with and without anconeus muscle flap for treatment of refractory lateral epicondylitis. *J Shoulder Elbow Surg*. 2015;24(2):236–241.
66. Buchbinder R, Johnston RV, Barnsley L, et al. Surgery for lateral elbow pain. *Cochrane Database Syst Rev*. 2011;(3): CD003525.
67. Grewal R, MacDermid JC, Shah P, et al. Functional outcome of arthroscopic extensor carpi radialis brevis tendon release in chronic lateral epicondylitis. *J Hand Surg Am*. 2009;34(5): 849–857.
68. Szabo SJ, Savoie FH 3rd, Field LD, et al. Tendinosis of the extensor carpi radialis brevis: an evaluation of three methods of operative treatment. *J Shoulder Elbow Surg*. 2006;15(6): 721–727.
69. Kwon BC, Kim JY, Park KT. The Nirschl procedure versus arthroscopic extensor carpi radialis brevis debridement for lateral epicondylitis. *J Shoulder Elbow Surg*. 2017;26(1): 118–124.
70. Han SH, Lee JK, Kim HJ, et al. The result of surgical treatment of medial epicondylitis: analysis with more than a 5-year follow-up. *J Shoulder Elbow Surg*. 2016;25(10):1704–1709.
71. Zonno A, Manuel J, Merrell G, et al. Arthroscopic technique for medial epicondylitis: technique and safety analysis. *Arthroscopy*. 2010;26(5):610–616.
72. Beyzadeoglu T, Bekler H, Gokce A. Skin and subcutaneous fat atrophy after corticosteroid injection for medial epicondylitis. *Orthopedics*. 2011;34(8):570.
73. Smith AM, Castle JA, Ruch DS. Arthroscopic resection of the common extensor origin: anatomic considerations. *J Shoulder Elbow Surg*. 2003;12(4):375–379.

74. Kalainov DM, Cohen MS. Posterolateral rotatory instability of the elbow in association with lateral epicondylitis. A report of three cases. *J Bone Joint Surg Am*. 2005;87(5):1120–1125.
75. Coleman B, Quinlan JF, Matheson JA. Surgical treatment for lateral epicondylitis: a long-term follow-up of results. *J Shoulder Elbow Surg*. 2010;19(3):363–367.
76. Lattermann C, Romeo AA, Anbari A, et al. Arthroscopic debridement of the extensor carpi radialis brevis for recalcitrant lateral epicondylitis. *J Shoulder Elbow Surg*. 2010;19(5):651–656.
77. Hobbs MC, Koch J, Bamberger HB. Distal biceps tendinosis: evidence-based review. *J Hand Surg Am*. 2009;34(6):1124–1126.
78. Rokito AS, McLaughlin JA, Gallagher MA, et al. Partial rupture of the distal biceps tendon. *J Shoulder Elbow Surg*. 1996;5(1): 73–75.
79. O'Driscoll SW, Goncalves LB, Dietz P. The hook test for distal biceps tendon avulsion. *Am J Sports Med*. 2007;35(11):1865–1869.
80. Mair SD, Isbell WM, Gill TJ, et al. Triceps tendon ruptures in professional football players. *Am J Sports Med*. 2004;32(2): 431–434.
81. Kijowski R, Tuite M, Sanford M. Magnetic resonance imaging of the elbow. Part II: Abnormalities of the ligaments, tendons, and nerves. *Skeletal Radiol*. 2005;34(1):1–18.
82. Chew ML, Giuffre BM. Disorders of the distal biceps brachii tendon. *Radiographics*. 2005;25(5):1227–1237.
83. Radunovic G, Vlad V, Micu MC, et al. Ultrasound assessment of the elbow. *Med Ultrason*. 2012;14(2):141–146.
84. Martinoli C, Bianchi S, Giovagnorio F, et al. Ultrasound of the elbow. *Skeletal Radiol*. 2001;30(11):605–614.
85. Vardakas DG, Musgrave DS, Varitimidis SE, et al. Partial rupture of the distal biceps tendon. *J Shoulder Elbow Surg*. 2001;10(4):377–379.
86. Sanli I, Morgan B, van Tilborg F, et al. Single injection of platelet-rich plasma (PRP) for the treatment of refractory distal biceps tendonitis: long-term results of a prospective multicenter cohort study. *Knee Surg Sports Traumatol Arthrosc*. 2016;24(7):2308–2312.
87. Durr HR, Stabler A, Pfahler M, et al. Partial rupture of the distal biceps tendon. *Clin Orthop Relat Res*. 2000;374:195–200.
88. Kelly EW, Steinmann S, O'Driscoll SW. Surgical treatment of partial distal biceps tendon ruptures through a single posterior incision. *J Shoulder Elbow Surg*. 2003;12(5):456–461.
89. Khan AD, Penna S, Yin Q, et al. Repair of distal biceps tendon ruptures using suture anchors through a single anterior incision. *Arthroscopy*. 2008;24(1):39–45.
90. Grewal R, Athwal GS, Macdermid JC, et al. Single versus double-incision technique for the repair of acute distal biceps tendon ruptures: a randomized clinical trial. *J Bone Joint Surg Am*. 2012;94(13):1166–1174.
91. Frazier MS, Boardman MJ, Westland M, et al. Surgical treatment of partial distal biceps tendon ruptures. *J Hand Surg Am*. 2010;35(7):1111–1114.
92. van Riet RP, Morrey BF, Ho E, et al. Surgical treatment of distal triceps ruptures. *J Bone Joint Surg Am*. 2003;85-A(10): 1961–1967.
93. Wolf JM, McCarty EC, Ritchie PD. Triceps reconstruction using hamstring graft for triceps insufficiency or recurrent rupture. *Tech Hand Up Extrem Surg*. 2008;12(3):174–179.
94. Kelly EW, Morrey BF, O'Driscoll SW. Complications of repair of the distal biceps tendon with the modified two-incision technique. *J Bone Joint Surg Am*. 2000;82-A(11):1575–1581.
95. Failla JM, Amadio PC, Morrey BF, et al. Proximal radioulnar synostosis after repair of distal biceps brachii rupture by the two-incision technique. Report of four cases. *Clin Orthop Relat Res*. 1990;253:133–136.
96. Anakwenze OA, Kancherla VK, Warrender W, et al. Outcomes of modified 2-incision technique with use of indomethicin in treatment of distal biceps tendon rupture. *Orthopedics*. 2011;34(11):e724–e729.
97. Cohen MS. Complications of distal biceps tendon repairs. *Sports Med Arthrosc*. 2008;16(3):148–153.
98. Katolik LI, Fernandez J, Cohen MS. Acute failure of distal biceps reconstruction: a case report. *J Shoulder Elbow Surg*. 2007;16(5):e10–e12.
99. Saini M, Canoso JJ. Traumatic olecranon bursitis. Radiologic observations. *Acta Radiol Diagn (Stockh)*. 1982;23(3A): 255–258.
100. McAfee JH, Smith DL. Olecranon and prepatellar bursitis. Diagnosis and treatment. *West J Med*. 1988;149(5):607–610.
101. McFarland EG, Mamanee P, Queale WS, et al. Olecranon and prepatellar bursitis: treating acute, chronic, and inflamed. *Phys Sportsmed*. 2000;28(3):40–52.
102. Weinstein PS, Canoso JJ, Wohlgethan JR. Long-term follow-up of corticosteroid injection for traumatic olecranon bursitis. *Ann Rheum Dis*. 1984;43(1):44–46.
103. Ogilvie-Harris DJ, Gilbart M. Endoscopic bursal resection: the olecranon bursa and prepatellar bursa. *Arthroscopy*. 2000;16(3):249–253.
104. Nussbaumer P, Candrian C, Hollinger A. Endoscopic bursa shaving in acute bursitis. *Swiss Surg*. 2001;7(3):121–125.
105. Rhyou IH, Park KJ, Kim KC, et al. Endoscopic olecranon bursal resection for olecranon bursitis: a comparative study for septic and aseptic olecranon bursitis. *J Hand Surg Asian Pac Vol*. 2016;21(2):167–172.
106. Stewart NJ, Manzanares JB, Morrey BF. Surgical treatment of aseptic olecranon bursitis. *J Shoulder Elbow Surg*. 1997;6(1): 49–54.
107. Degreef I, De Smet L. Complications following resection of the olecranon bursa. *Acta Orthop Belg*. 2006;72(4):400–403.